

MONA, United Kingdom

WMO: 033010

Lat: 53.2600N

Lon: 4.3761W

Elev: 62

StdP: 100.58

Time Zone: 0.00 (EUW)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.3	-1.7	-5.7	2.4	-1.2	-4.3	2.7	-0.3	18.6	8.6	16.9	8.7	1.9	60	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	22.7	17.2	20.7	16.4	19.3	15.7	18.1	20.9	17.2	19.4	16.5	18.3	4.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.3	18.9	16.4	11.7	18.0	15.7	11.3	17.2	51.3	20.9	48.7	19.5	46.5	18.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.4	12.0	-5.9	26.4	3.0	2.2	-8.1	27.9	-9.8	29.2	-11.5	30.5	-13.7	32.1		
(5)			-6.3	20.3	3.0	1.5	-8.5	21.3	-10.2	22.2	-11.8	23.1	-14.0	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	5.4	5.1	6.1	8.2	10.8	13.4	15.1	14.7	13.1	11.0	7.5	6.3		
	DBStd	4.38	2.78	2.75	2.62	2.51	2.47	2.11	2.04	1.71	2.15	2.64	2.92	3.31		
	HDD10.0	708	143	137	123	64	18	1	0	0	2	20	83	117		
	HDD18.3	3132	401	369	379	303	234	148	102	113	156	228	324	373		
	CDD10.0	622	1	1	2	12	42	104	158	145	96	51	9	3		
	CDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0		
	CDH23.3	34	0	0	0	0	3	11	17	1	1	0	0	0		
	CDH26.7	2	0	0	0	0	0	1	2	0	0	0	0	0		
(14) Wind	WSAvg	5.5	6.0	6.1	5.6	5.2	5.4	5.0	4.9	5.3	5.2	5.6	5.3	6.4		
(15) Precipitation	PrecAvg	845	80	59	60	54	51	53	56	70	74	96	104	93		
	PrecMax	1055	159	161	152	103	127	129	127	141	160	209	220	254		
	PrecMin	626	7	7	13	7	10	11	9	11	3	19	27	17		
	PrecStd	99	36	31	29	25	27	29	27	27	37	47	40	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	13.6	17.4	20.8	22.9	25.0	25.2	22.2	22.1	18.8	14.7	13.2		
		MCWB	11.3	9.2	11.4	14.2	17.1	17.3	19.0	17.3	17.8	15.9	13.0	11.8		
	2%	DB	10.7	10.6	12.6	16.6	19.0	21.3	23.1	20.0	19.7	16.6	13.8	12.1		
		MCWB	10.3	9.1	9.9	11.9	14.9	16.5	17.6	16.4	16.0	14.8	13.2	11.6		
	5%	DB	9.9	9.8	10.9	14.0	16.8	19.2	20.9	18.7	18.0	15.7	12.9	11.4		
		MCWB	9.4	9.0	9.3	10.5	13.3	15.2	16.6	15.8	15.7	14.5	12.3	10.9		
	10%	DB	9.4	9.1	9.8	12.3	15.2	17.6	19.3	17.8	16.9	14.9	12.1	10.7		
		MCWB	8.8	8.5	8.4	9.7	12.3	14.4	16.0	15.4	14.8	13.9	11.3	10.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	10.5	11.9	14.4	17.8	18.9	19.9	18.1	18.2	16.2	14.1	12.5		
		MCDB	11.6	11.7	15.3	19.5	21.6	23.1	24.1	19.8	20.5	17.4	14.3	12.6		
	2%	WB	10.5	9.8	10.5	12.4	15.7	17.1	18.4	17.2	17.1	15.6	13.4	11.8		
		MCDB	10.6	10.3	12.0	16.0	18.3	20.2	21.7	18.8	18.4	16.3	13.6	11.9		
	5%	WB	9.6	9.2	9.6	11.1	13.9	15.8	17.4	16.6	16.1	14.9	12.5	11.1		
		MCDB	9.8	9.6	10.8	13.0	15.9	18.2	19.8	18.0	17.3	15.5	12.8	11.3		
	10%	WB	9.0	8.6	8.9	10.3	12.9	14.9	16.5	16.1	15.4	14.0	11.4	10.3		
		MCDB	9.3	9.0	9.7	11.9	14.4	17.0	18.5	17.3	16.5	14.7	12.0	10.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	6.4	7.1	7.1	7.2	7.0	6.3	6.7	5.7	5.5	4.7		
		MCDBR	4.2	5.3	7.8	10.3	9.9	10.8	10.3	7.7	8.8	5.9	5.2	4.7		
	5% WB	MCWBR	4.7	4.4	5.6	6.2	6.4	6.8	6.1	5.1	5.9	4.5	5.1	5.2		
		MCWBR	4.3	4.6	6.9	8.6	8.1	8.7	8.5	6.4	6.7	5.0	4.8	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.343	0.368	0.386	0.391	0.383	0.386	0.371	0.364	0.350	0.327	0.320		
		taud	2.310	2.377	2.339	2.317	2.356	2.404	2.401	2.456	2.471	2.450	2.376	2.286		
	Ebn at Noon		630	757	812	844	858	868	858	852	813	740	635	563		
		Edh at Noon	62	78	98	114	115	111	109	98	86	72	57	53		
(44) All-Sky Solar Radiation	RadAvg		0.65	1.36	2.59	4.09	5.19	5.52	5.04	4.22	3.01	1.67	0.81	0.48		
	RadStd		0.06	0.12	0.19	0.29	0.36	0.47	0.44	0.28	0.19	0.15	0.08	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page